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Why not both? The effects of innovation and capital on productivity in New Zealand

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Executive summary

New Zealand has not seen the same growth in productivity as comparable countries. Increased capital intensity and higher levels of innovation are pathways to greater productivity growth that are closely connected. Investment in more technologically sophisticated capital could contribute to productivity improvements along both pathways at once.

New Zealand has relatively low capital intensity, with the high cost of capital a contributing factor. Though New Zealand’s investment rate has tracked with other advanced economies, it has not kept pace with rising labour utilisation, leaving the country capital shallow.

Innovation increases an economy’s productivity by allowing for a more efficient mix of capital and labour. Improved technology is one route to greater innovation. Technology can be created anew or adopted from elsewhere. Neither creation nor adoption is unambiguously preferable from a productivity perspective. The optimal combination of creation and adoption in supporting productivity depends on context. We set out a number of reasons why adoption through capital investment may be an important path to innovation in New Zealand.

New Zealand creates technology at a lower rate than comparable economies, in part because its Research and Development (R&D) expenditure is lower. New Zealand also struggles to convert R&D activity into outputs and broader productivity benefits. Although some business R&D appears an exception to this rule, New Zealand-specific evidence is limited, which makes it difficult to be definitive about the productivity benefits of R&D.

¹ The authors would like to thank Rebecca McDonald (formerly of the Treasury) who developed much of the framing around innovation creation and diffusion in previous internal work. Thanks to Renee Philip, Harry Nicholls, Ken Tsang, Nick Robertson as well as other colleagues from the Treasury and the Ministry of Business Innovation and Employment for constructive comments. An early version of this paper was presented to the June 2025 New Zealand Association of Economics conference and has benefitted from helpful questions and comments from the audience.

Adopting and adapting new technologies, including through capital investments, typically costs less than creating them. Adopting new technologies is shown to have strong positive effects on productivity, and could potentially benefit a large share (up to 95%) of New Zealand firms. Given New Zealand's struggle to create technology, greater adoption from overseas could form a crucial part of the country's optimal approach to innovation. At the same time, a dynamic and, to some extent, complementary relationship may exist between creation and adoption – suggesting stronger adoption of new technology may enhance domestic scientific activity.

Despite adopting some general-purpose digital technologies at a pace similar to comparable countries, there are signs that technology diffusion (the aggregate economy-wide rate of firm-level technology adoption) is low and slowing. To better understand opportunities for increasing diffusion, we capture in a framework the key factors that affect diffusion and their links to capital investment. We organise the framework around three firm-level factors: exposure and access to new technology, incentives to adopt new technology, and capacity to adopt new technology.

We find there are four common channels that directly affect technology diffusion and capital intensity: importing, foreign investment, input costs, and access to finance. New Zealand appears, based on initial assessment, to be weaker than the OECD average across all four channels. Improvements on these channels and across framework settings (such as, competition), could boost productivity growth by lifting both capital intensity and innovation.

The fundamental implication is that policy focussing exclusively on technology creation may miss a key pathway to greater innovation: diffusion through new capital investment. Equally, policies exclusively focussed on capital intensity may miss the role of investment in increasing innovation and lifting the technological sophistication of the capital stock.

1 Introduction

New Zealand's productivity growth has been weak for several decades. As a result, the current level of productivity remains below that in many comparator countries. Over recent years the Treasury and others have looked at the potential broad drivers of productivity performance, including: macroeconomic imbalances, low capital intensity, small domestic markets, distance from overseas markets, the slow diffusion of productivity-enhancing change, and the role of services and international connections. Further work has assessed New Zealand's long-term productivity performance in the context of the global productivity slowdown, and its implications for Treasury's forecasts and fiscal projections.²

This paper fills an analytical gap regarding the connections between productivity, capital investment, innovation and technology. Often innovation and capital investment are treated separately. That is, innovation is thought of as technology creation in the sense of discovering new ideas and inventing new products. Capital is often thought of in its role as a factor of production – quantitatively, and agnostic of its technological qualities. By looking closely at technology diffusion (the economy-wide process where firms adopt new technologies) this paper highlights the complex relationship between capital and innovation, and explores the implications for New Zealand's productivity.

Innovation can be defined more broadly than purely creation. Seen in this way, innovation exists on a spectrum. At one end lies the invention of novel, new-to-world ideas, products and processes (eg, a bio-tech break through). At the other extreme lies the diffusion, adoption and adaptation of existing technologies (eg, the development of smartphone apps, or production line upgrades). Iterative improvement of existing technology falls in the middle of the spectrum. In this paper we use 'adoption' to refer to the uptake of new technologies by any individual firm, and 'diffusion' to refer to the process at an economy- or industry-level where many firms adopt new technologies.

Technology diffusion is far from a novel area of study. Economists have for decades analysed rates of technology diffusion within and between countries (Romer, 1994). Interest in diffusion in advanced economies has grown of late, owing to four factors:

- 1 Support for the hypothesis that a weak 'diffusion machine', which fails to support the economy-wide adoption of technologies and best practices, may partly explain the productivity slowdown in advanced economies (Andrews, Nicoletti, & Timiliotis, 2018). For example, the Australian Productivity Commission (APC) (2023) estimates reviving diffusion could lift Australia's Gross Domestic Product (GDP) by between 1.6% and 4.9%.
- 2 The potential broad-based benefits of technology diffusion. The APC (2023) estimates that for 98% of Australian firms, innovation is about adopting technology rather than creating it. In New Zealand the split seems similar with 95% of firms appearing to innovate through adopting and adapting technologies.³

² For a selection of previous Treasury research and perspectives, see: Treasury (2014); Conway (2016); Janssen, Galt & Bollinger (2022) Galt (2023); and Treasury (2023). On forecasts and fiscal considerations, see: Treasury (2024). For other perspectives, see: NZPC (2023); OECD (2024); and Conway (2025).

³ According to Business Operations Survey (BOS) data from 2018-2023. BOS is an annual survey of New Zealand business and covers a sample of approximately 50,000 firms with 6 or more employees.

- 3 Technology diffusion is a necessary building block for later stages of innovation. As Breznitz (2021) and Ridley (2020) note, most of the benefits accruing from innovation come about through incremental refinements to products and ideas that already exist but have some as-yet unrealised economic potential. Because incremental refinement only works if there is something – a product or idea – to be refined in the first place, this makes diffusion a critical precursor and complement to further innovation.
- 4 An expectation that digitisation, electrification, and use of Artificial Intelligence (AI) could rekindle growth – an expectation that itself assumes widespread diffusion of AI and complementary technologies (Mukherjee & Nicholls, 2024). Estimates of AI's economic impact vary widely, though most are strongly positive (APC, 2025).

The Treasury has previously identified technology diffusion as a potential roadblock for New Zealand's productivity (Makhlouf, 2015; Treasury, 2024), including in relation to AI (Mukherjee & Nicholls, 2024). In our judgement, the issue warrants continued attention.

Despite its economic significance, technology diffusion often receives secondary consideration in the public discussion of science and technology, where notions of discovery and invention tend to hold sway. A narrative that emphasises breakthrough moments of discovery creates the risk of people thinking about innovation in an overly narrow way.

Given that New Zealand performs less strongly than other countries both on capital intensity and measures of innovation, improved technology diffusion – which often involves capital investment – offers a way to increase both capital intensity and innovation. To better understand technology diffusion we develop a framework that captures the key factors and channels shaping firm decisions to adopt new technology.

The paper proceeds as follows. Section 2 defines key concepts. We then show how New Zealand's poor productivity performance is tied to capital intensity and to innovation. Section 3 lays out the evidence for the country's low capital intensity and relatively low rates of technology creation and diffusion. Section 4 outlines a framework for understanding technology diffusion, examines the factors and channels for technology diffusion and links to capital intensity, and offers an initial assessment of the channels' strengths. Finally, we suggest areas for future work.

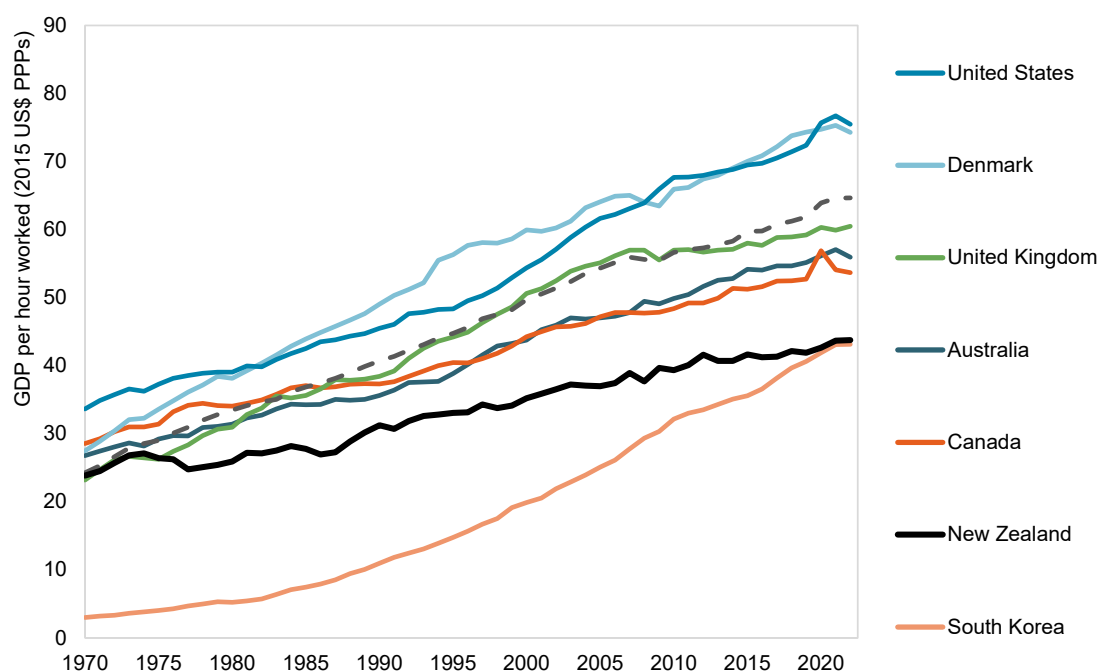
2 Pathways to productivity growth

Material living standards are largely determined by productivity growth. New Zealand's labour productivity (measured as GDP per hour worked) began to diverge from OECD peers in the 1970s and has since endured 50 years of relative decline. Despite some modest growth after the early 1990s, the New Zealand economy remains at a lower level of GDP per hour worked than OECD peers. New Zealand has not shown signs of sustained catch-up (Figure 1).⁵

Growth accounting and the role of productivity

Long-term economic output is a function of capital inputs, labour inputs, and Multi-Factor Productivity (MFP). The economy's aggregate production function can be represented by a growth accounting framework that expresses how these inputs and factors combine.⁶ The production function reflects key macroeconomic relationships: the complementarity of labour and capital inputs, how increases in these inputs increase GDP per hour worked (all other factors being equal), and how diminishing returns affects both labour and capital inputs.

Figure 1: GDP per hour worked, 2015 US\$ PPPs



Source: OECD Productivity Database.

MFP is a concept that captures how efficiently actors in the economy combine labour and capital to produce a unit of output. MFP cannot be measured directly, unlike labour and capital. Rather, it is measured as a residual – that part of growth not directly attributable to shifting levels of labour or capital.

⁵ New Zealand's more recent income growth since the mid-1990s has been supported by labour productivity growth, an increasing share of the population in employment and a higher terms of trade (Galt, 2023).

⁶ Expressed formally as $Y = f(K, L, MFP)$ described by a Cobb-Douglas production function, or similar.

MFP encompasses the economy's level of innovation and technology, management practices, organisational changes, network effects, economies of scale, market competition, as well as statistical errors. As measurement of labour and capital inputs improve (by including some intangible forms of capital, for example), less of other factors' contributions to output is accounted for in the MFP residual. Labour productivity – the amount of economic output per unit of labour input, Y/L – is affected by both MFP and capital intensity (the ratio of capital to labour, K/L). This type of production function, with its two factors of production (capital and labour) is simplified. It does not, for example, fully account for other forms of capital, most notably human capital and intangible capital.

Its simplicity notwithstanding, the growth accounting framework described above suggests potential causes of New Zealand's productivity performance could include low capital intensity and low MFP. It also implies a link from MFP to capital intensity. Namely, a positive shock to MFP (eg, through the widespread diffusion of new technology) increases both output and, after adjustment, capital intensity. The intuition is that a more innovative and productive economy offers a greater number of investible opportunities with higher returns (André & Gal, 2024; Janssen, Galt, & Bollinger, 2022). Establishing the relative importance of MFP and capital intensity for growth is subject to measurement challenges around capital and the form of the production function.

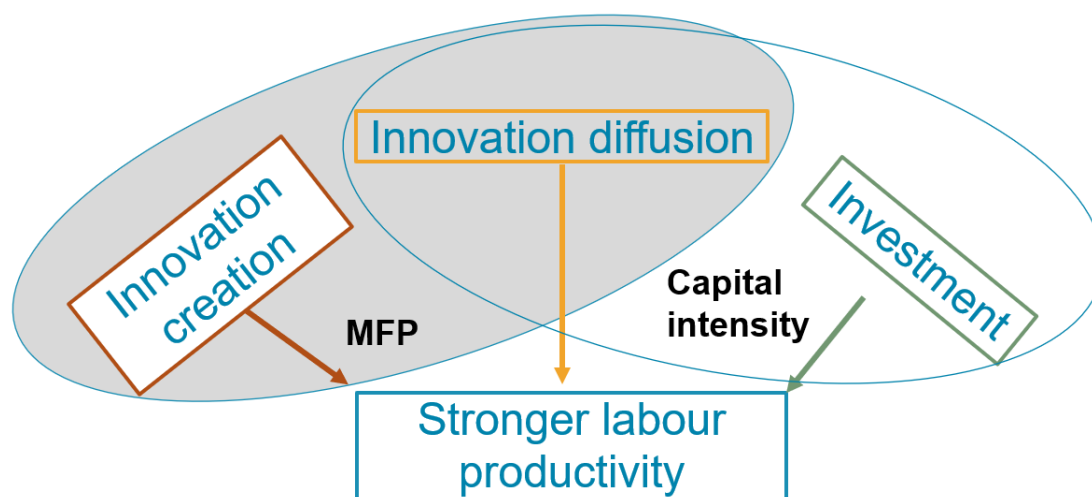
The cross-cutting effects of innovation on measured productivity

Innovation is the invention and introduction of new or improved consumer and capital products, and processes (IMF, 2024). It “means much more than invention, because [innovation] implies developing an invention to the point where it catches on because it is sufficiently practical, affordable, reliable and ubiquitous to be useful” (Ridley, 2020, p. 15).

Technology is a key route to innovation and economic growth. The creation of technology is one approach. The adoption and diffusion of technology through investment in more sophisticated capital is another (somewhat overlooked) approach. Neither approach is unambiguously preferable from a productivity perspective. The source of a technology has little bearing on its economic effects. As Hausmann (2017) notes: “fundamentally technology is a way to transform ‘the world as I found it’ into ‘the world as I want it to be’ – from pastures to milk, from soybeans to chicken tenders, from silicon to smartphones. And it depends on three forms of knowledge: embedded knowledge in tools; codified knowledge in recipes, manuals, and protocols; and tacit knowledge, or knowhow, in brains”. Embedded knowledge is physical capital; codified knowledge is intangible capital; and tacit knowledge is human capital or labour.

The economic effects of technology diffusion may not be straightforwardly captured by growth accounting frameworks. Statisticians rely on price and cost information to capture (in the measure of capital inputs) the knowledge embedded in tools and written down in manuals (Corrado, Goodridge, & Haskel, 2011). But price and cost information are unlikely to fully account for levels of knowledge and technological sophistication, nor for any network effects associated with widespread technological change. Residual productivity changes resulting from technological change will be captured in MFP. As a result, the productivity effects of technology diffusion may arise across both capital and MFP (Figure 2).

Figure 2: The cross-cutting effects of technology on measured productivity



Source: The Treasury, 2025.

The critical real-world insight of the preceding discussion is that the relationships between capital, labour, and MFP are complex and interconnected. Each offers pathways to increased innovation.⁷ Accordingly, this paper focusses on the connections between technology diffusion, innovation and capital intensity. This focus does not imply that all increases in capital enhance innovation or vice versa. Not all capital investment will embed or codify new technology (eg, when capital is replaced like-for-like) and not all technology is embedded in tangible capital or codified in intangible capital (eg, tacit knowhow is embodied in labour inputs).

⁷ Technology and capital are not the only pathways to greater productivity. Human capital is also important. The Treasury has recently completed a separate work programme on human capital, including analysing historical trends in human capital development in New Zealand (Reid, 2025), reviewing the evidence about the outcomes from human capital investment, taking into account New Zealand's characteristics (Schaer, 2025), and developing future scenarios for lifting human capital (Ng, 2025).

Firm-level analysis and the importance of innovation

Private firms are the main actors in the economy. They generate around 70% of New Zealand's GDP,⁸ and take most of the decisions and implement many of the changes that lead to innovation. The aggregate economic shifts described by growth accounting reflect developments at the level of the firm. It is key therefore to look at issues of innovation, capital and productivity from a firm perspective.

Firms may upgrade their technology by adopting and adapting existing ideas and technologies, or as the result of inventing and operationalising new technologies and processes. In taking decisions on whether and how to adopt new technology, firms draw on price signals and market information, and are incentivised to maximise profitability and future ability to do business. Firms at the global frontier may seek to invent new technologies to stay ahead of the market. Those behind the frontier may try to catch up by adopting pre-existing technology (NZPC, 2021). As we outline in Section 4, a firm's decision to adopt new technology is influenced by its access and exposure to new technologies, its incentive structure and its capacity.

The OECD emphasises the role of firm-level innovation, diffusion and reallocation in driving aggregate productivity (André & Gal, 2024; Andrews, Criscuolo, & Gal, 2015). These firm-level processes improve overall allocative efficiency. They lead to the expansion or contraction of incumbents, and to the entry of higher productivity firms – resulting ultimately in resource reallocation towards more productive firms. More recently, the OECD linked these firm-level processes to firm capabilities across capital, labour and MFP, and to public policy (André & Gal, 2024). Overall, dynamics at the firm level support the conclusion drawn from aggregate growth accounting that innovation and investment both promote productivity growth.

⁸ Estimated as GDP less Core Crown expenses (Treasury, 2025).

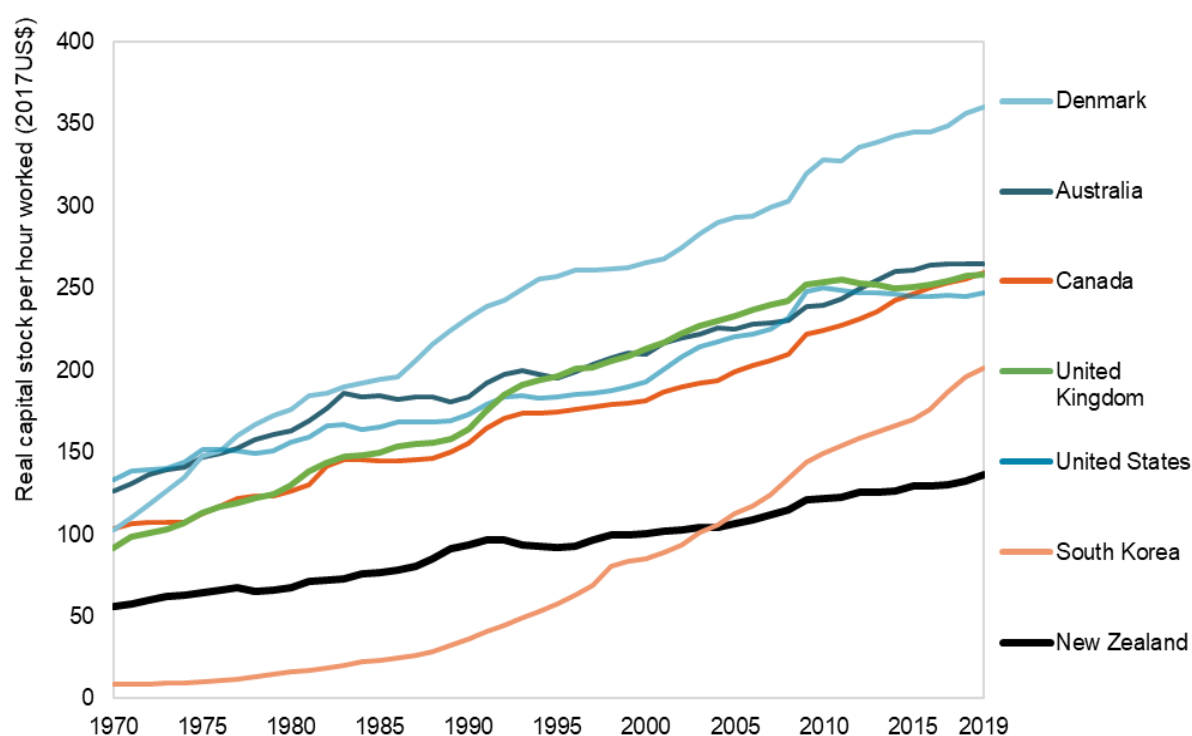
3 Capital and innovation in New Zealand

New Zealand has low capital intensity

New Zealand began the 1970s with a low level of capital intensity (capital per hour worked) and has enjoyed relatively strong growth in labour utilisation rates since. As a result, New Zealand's investment rate would need to have accelerated over many years to converge with the average OECD economy's capital intensity. It has not made this acceleration, and New Zealand's current overall capital intensity is around 50% of OECD comparators (Figure 3).⁹

New Zealand's relatively low capital intensity and middling investment rate should both be seen in the context of a rise in service industries (which tend to be capital light) and a generalised 20-year investment slowdown across advanced economies. Rates of investment across the OECD have barely caught up with the trend prior to 2008 (Dlugosch, Glanville, Hooley, Ozturk, & Westmore, 2025; Hanappi, Millot, & Turban, 2023), and the contribution of capital per worker to GDP growth declined considerably as a result.

Figure 3: Capital intensity, New Zealand and selected OECD countries



Source: Penn World Tables. See: Inklaar, Woltjer and Albarran (2019); Gouma and Inklaar (2022); and Gouma, Inklaar and Marapin Gouma (2023).

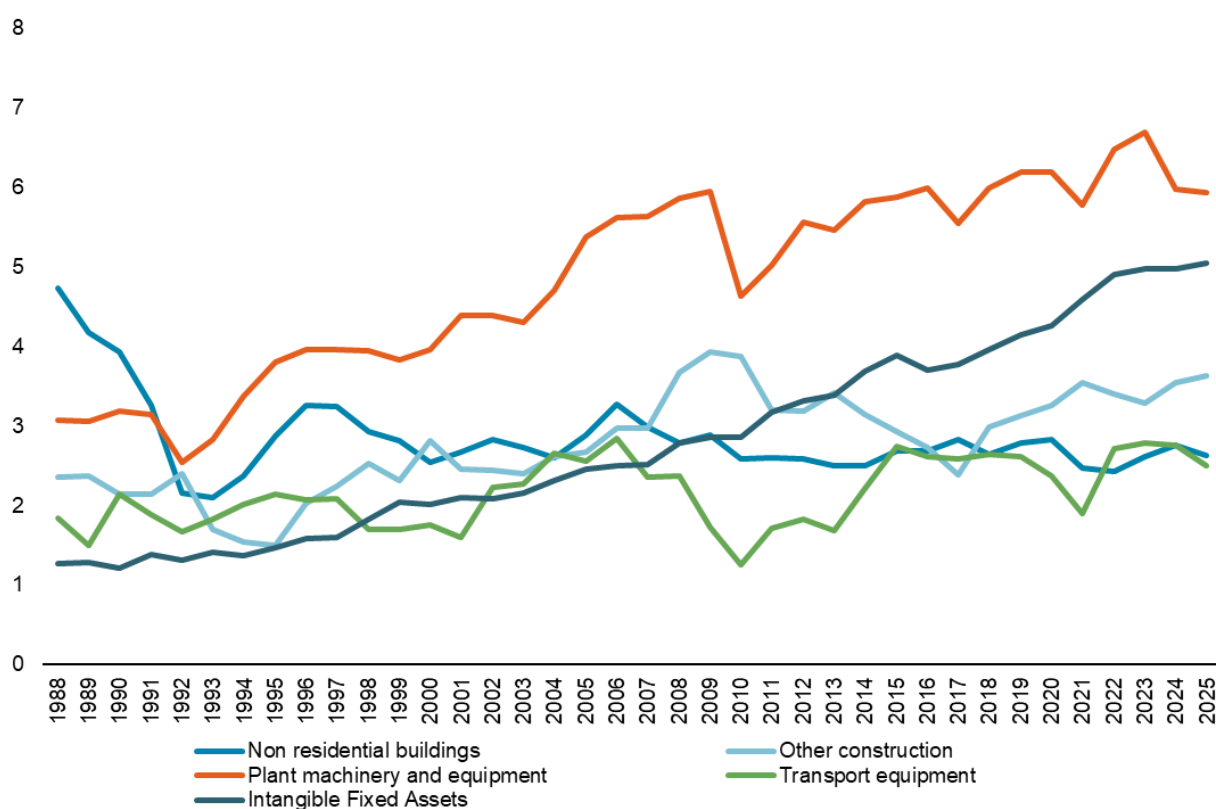
Note: Capital stock series comprises nine asset types: residential buildings, other structures, information technology, communication technology, other machinery, transport equipment, software, other intellectual property products, and cultivated assets (such as livestock for breeding and vineyards).

⁹ This figure depends on how capital is measured (Janssen, 2018; Janssen, Galt, & Bollinger, 2022).

Multiple factors drive depressed capital intensity (Conway, 2016; Pells, 2020; Treasury, 2024). New Zealand's high cost of capital (and consequently low returns relative to other economies) is a persistent dampener on business investment.¹⁰ Structural and cyclical factors, including low aggregate demand, policy uncertainty and weaker competitive pressures, may also have a hand in curbing investment in OECD countries (Dlugosch, Glanville, Hooley, Ozturk, & Westmore, 2025; IMF, 2025)

As well as a lower level of capital, the composition of New Zealand's investment differs from other countries. New Zealand has a comparatively high rate of residential investment, reflecting both population pressures and higher relative returns. Within business investment, we see an increase in plant, machinery and equipment investment in real terms, and an increase in intangibles and other construction (Figure 4). We cannot attribute all the greater investment in plant, machinery and equipment to capital upgrading and technology diffusion – since some of it will reflect replacement costs. Separately, the growing share of intangibles reflects a rise in intangible-heavy businesses across the economy. (Intangible capital includes, for example, Intellectual Property, software, marketing assets, and brand equity.)

Figure 4: Components of business investment, New Zealand, percent GDP



Source: Stats NZ National Accounts.

¹⁰ New Zealand's capital cost has been comparatively high for decades (Dupuy & Beard, 2008). Barriers to some groups and types of businesses accessing capital, barriers to foreign investment, high input prices of imported capital, and capacity constraints also drive cost (Inland Revenue, 2022).

New Zealand creates less technology than its peers

Alongside low capital intensity, slow productivity growth may be in part related to New Zealand's relatively weak record of creating new technologies. Conceptually, technology creation enhances productivity growth by generating knowledge, ideas and technologies that are to an extent non-rival, non-excludable and accumulative. The outputs of technology creation are, in short, difficult for their creators to control. Knowledge created by one firm can spill over in various ways to others, increasing overall productivity (Griliches, 1991). The tendency of knowledge to spill over means private actors may not invest in innovation to a socially optimal level. This market imperfection provides a (non-decisive) rationale for government intervention and helps explain public investments in R&D across the OECD.¹¹

In addition to spillovers, creating new technology and Intellectual Property (IP) can also support productivity growth by fostering internationally competitive firms operating at (or near) the global technology frontier. In theory, new technological offerings allow these firms to capture additional value, through mechanisms such as competitive differentiation and licensing (OECD, 2023).

The nature of technology creation means that it defies direct measurement. In what follows we take a conventional, however imperfect, approach to measuring technology creation by analysing both inputs (R&D expenditure) and outputs (patent filing activity). We also highlight some of the limitations with these metrics.

New Zealand has relatively low R&D expenditure, particularly among businesses

New Zealand spends less on R&D overall than the OECD average. Gross Expenditure on R&D (GERD) amounted to NZ\$6.4bn in 2024, accounting for approximately 1.5% of GDP. New Zealand's R&D intensity has risen since 2018, although it remains around half the OECD average (Figure 5). To put this in perspective, Finland – a country with many similar economic attributes to New Zealand – set a goal of increasing R&D intensity to 4% of GDP by 2030. As of this writing, Finland spends approximately 3%.

New Zealand has starkly lower Business Expenditure R&D (BERD) than other OECD economies, notwithstanding steady increases in recent years. New Zealand's BERD currently constitutes around 1% of GDP – up from 0.5% in 2014 but less than the 2% that the median OECD member spends. At the same time, public R&D (the government expenditure component in particular) makes up a comparatively large share of overall New Zealand GERD.

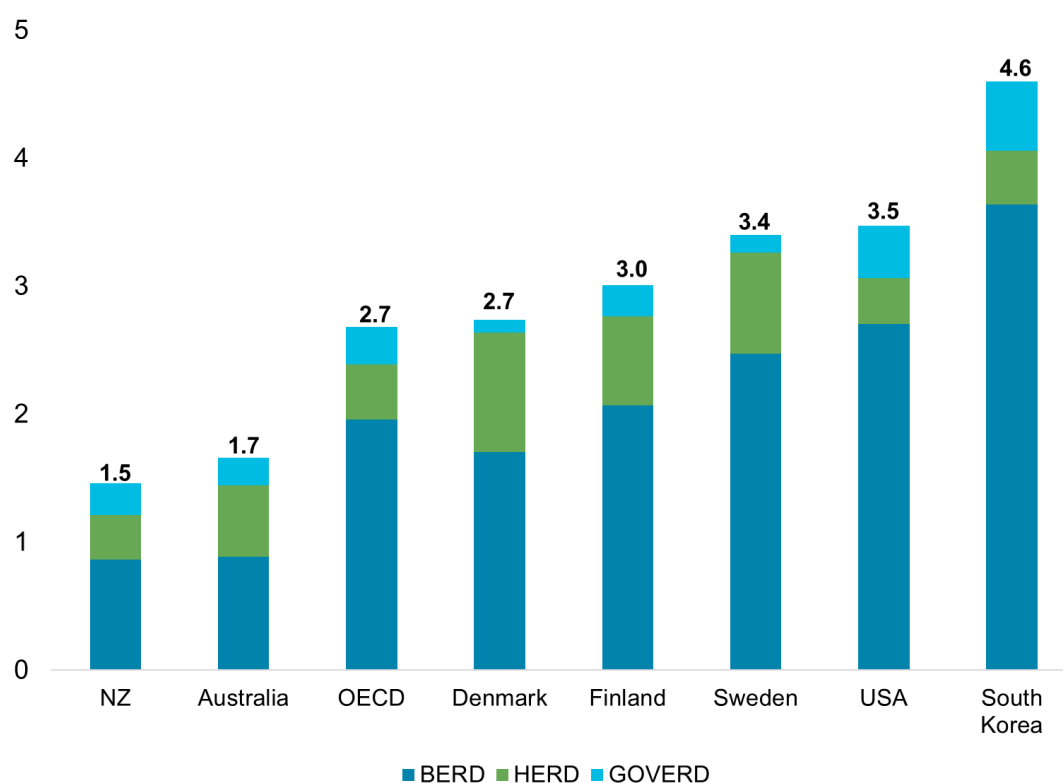
The sector composition of R&D in New Zealand is changing. While participation and spending on R&D has risen in most areas and on most measures¹² since 2010, growth in R&D on health and on services has outpaced growth elsewhere. As a result, gross R&D expenditure in 2024 for the purposes of health, information services and primary industries were each individually larger than gross R&D spending on manufacturing, which as recently as 2020 was the single largest expenditure category.

¹¹ The case for intervention on technology diffusion is less clear, though diffusion of some new technologies may be subject to information and risk asymmetries (Jaffe, Newell, & Stavins, 2005).

¹² Other measures of R&D activity include share of participating firms and number of R&D workers.

As the Treasury (2023) noted previously, public and private spending on R&D varies by industry (Figure 6). Government R&D (GOVERD) goes largely into the environment and to primary industries, which together make up nearly 75% of spending. BERD occurs more evenly across sectors, with sizeable expenditure into manufacturing and information and communications services. Health represents a major higher education R&D (HERD) outlay, with manufacturing and primary industries receiving less.

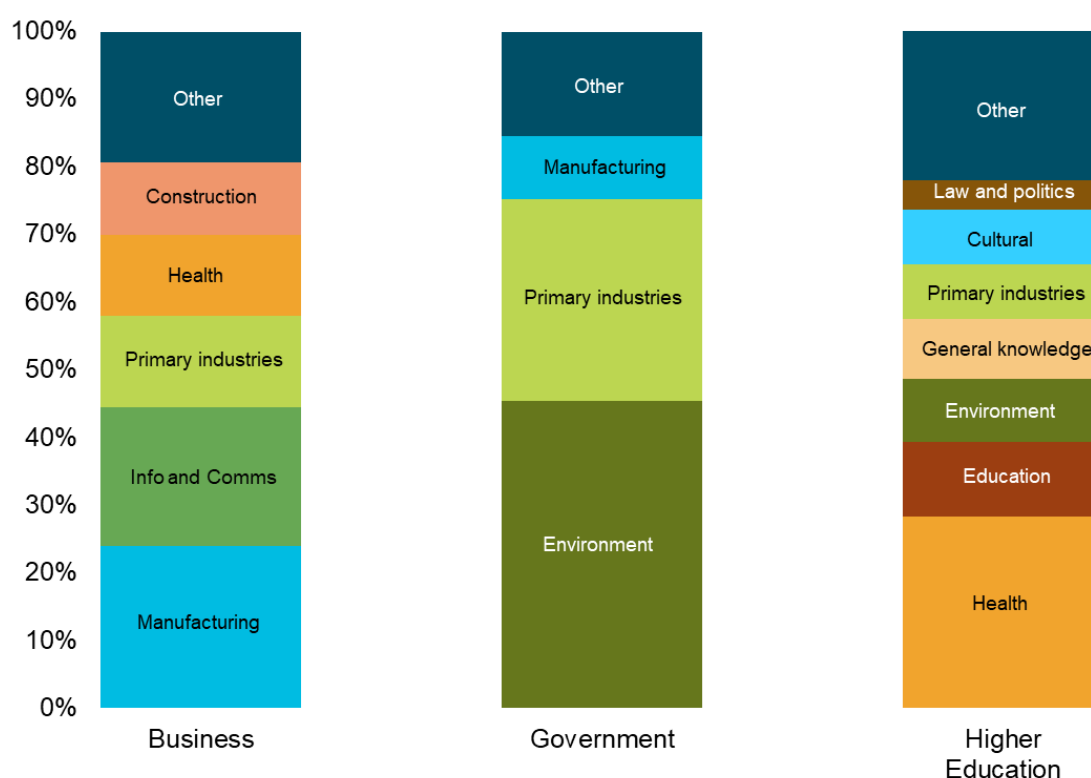
Figure 5: Gross expenditure on R&D, percent GDP, 2021



Source: OECD Main Science and Technology Indicators.

Note: BERD = Business Expenditure R&D, HERD = Higher Education Expenditure R&D, GOVERD = Government Expenditure R&D. BERD + HERD + GOVERD = GERD Gross Expenditure R&D.

Figure 6: R&D expenditure by purpose industry, New Zealand, 2020



Source: Stats NZ Research and Development Survey 2020.

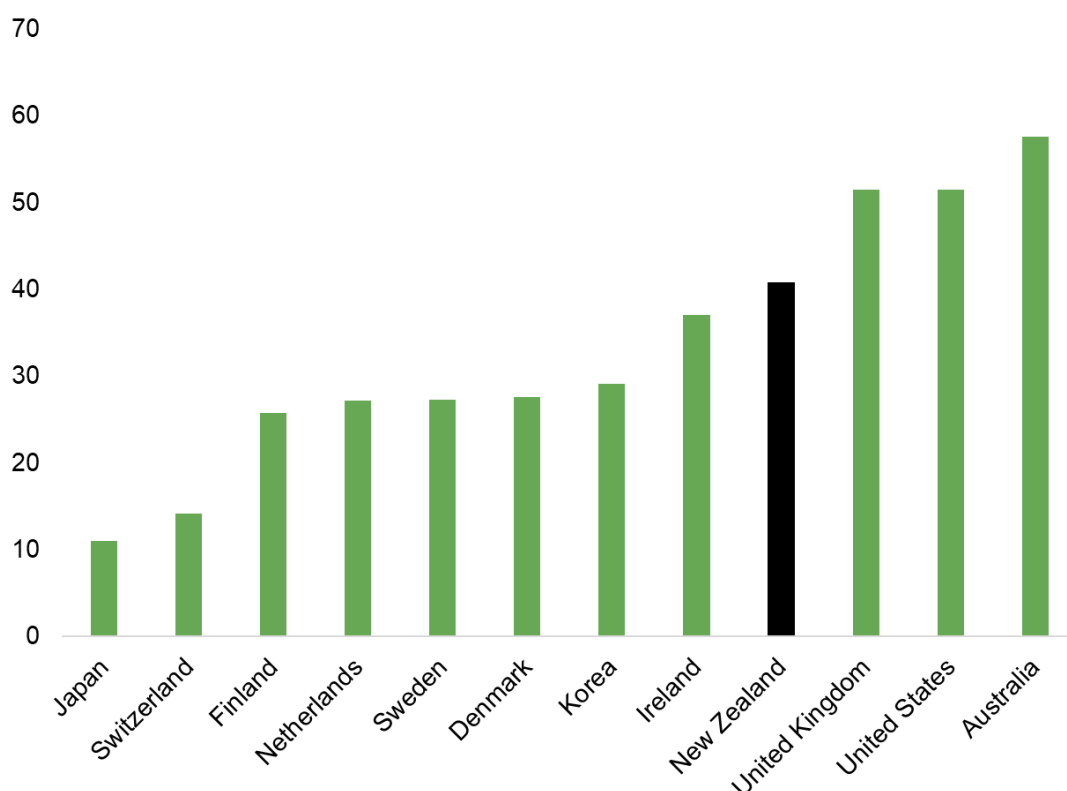
Outputs from R&D slightly lower than similar OECD countries

New Zealand has a strong record of inventiveness and pockets of excellence in some research areas, such as food technology (eg, growing golden kiwifruit). Nevertheless, it lags similar OECD countries and small advanced economies (SAEs) on most measures of technology output, including patenting IP. New Zealand's patenting performance improves when controlling for GDP and GERD (the latter is shown in Figure 7). In fact, on the ratio of patents to GERD, New Zealand outperforms some peers, including Australia. Yet the implied cost in R&D dollars for each high-quality patent filing is greater in New Zealand than in other SAEs – such as Finland and Denmark. New Zealand's R&D output, including IP filings, has been in decline since 2005 – despite increased inputs (Wakeman & Le, 2015). This decline is mirrored across the OECD (André & Gal, 2024).

Economists often use patent-based metrics to proxy for innovation. Such metrics, including those that we employ in this paper, have well-understood flaws (Griliches, 1998). These include the fact that pursuit of IP protection reflects sectoral features and business strategy as much as successful R&D (Appelt, Bajgar, Criscuolo, & Galindo-Rueda, 2022). In addition, the economic value of patents varies considerably, and some forms of highly beneficial innovation cannot be easily patented (eg, refinements to existing practice).

Anecdotally, New Zealand has a cultural inclination to tinker, a large primary sector, and rising economic and R&D activity in low-patent sectors, especially services (Hendy, 2013). These features could lead to the conclusion that patent metrics poorly describe innovation in New Zealand. But the current evidence suggests that patents correlate well with other indicators of innovation among New Zealand firms (Wakeman & Le, 2015). We take this to mean that patenting can tell us something valid about innovation in New Zealand.

Figure 7: Implied cost of triadic patent families filed, USD\$ 2021



Source: OECD Main Science and Technology Indicators, Treasury calculations.

Note: Implied cost of patents derived from USD\$ millions of GERD divided by number of triadic patent families filed. Implied costs do not reflect direct costs of patents, rather indicate relative R&D resource allocation.

In short, patent activity suggests New Zealand gets a middling level of output from its R&D expenditure. But what ultimately matters are the economic benefits generated by the search for, and creation of, new technology and ideas. In theory, these benefits could differ from those implied by patent activity measures. Benefits not captured by patenting include resilience and sustainability improvements that result from R&D. Such improvements could enhance the country's economic performance, broadly defined. R&D could, for example, support the sustainable use of water or reduce on-farm greenhouse gas emissions. The current paper's scope focusses on productivity and excludes consideration of these and other aspects of economic performance.

Overall, the evidence of productivity improvements from R&D in New Zealand paints a mixed picture. Among other issues, the evidence suffers from being increasingly out of date. In one area at least, however, the findings seem undeniable – private benefits. International studies consistently find that firms receive private benefits from R&D, and existing evidence indicates New Zealand firms are not an outlier in this regard (Fabling & Grimes, 2007; Johnson, Razzak, & Stillman, 2007).

Notwithstanding private benefits from R&D in New Zealand, the justification for government intervention in R&D rests in part on the grounds that it generates positive externalities in the form of productivity spillovers (Bloom, Van Reenen, & Williams, 2019). Evidence in relation to productivity spillovers from New Zealand-based R&D offers less-than-conclusive findings. Such as it is, the evidence can be summarised as suggesting that BERD increases productivity in other parts of the economy, while public R&D (GOVERD plus HERD) tends not to increase productivity in other parts of the economy.

A closer look at the evidence reveals certain conditions under which R&D generates spillovers. Johnson, Razzak and Stillman (2007) find that New Zealand BERD does spill over to firms in the same industry, and BERD in some industries (building, forestry and some services) increases wider economic productivity. These findings are consistent with research into agricultural R&D (Hall & Scobie, 2006) and match patterns identified in past research (Johnson, 2000). But more recent (albeit provisional) economy-wide data on New Zealand BERD (OECD, 2023) seem to contradict both Johnson et al (2007) and Johnson (2000). These new data indicate sizeable R&D spillovers, both within the same industry and to others.

In terms of public R&D, Johnson et al. (2007) find no evidence of spillover effects in New Zealand. Johnson et al.'s conclusion diverges somewhat from consistent international evidence that public R&D generally generates spillovers, though it aligns with a broader theme in the literature: that spillover effects from public R&D vary by sector and by country (Fieldhouse & Merten, 2023; Moretti, Steinwender, & Van Reenen, 2025).

New Zealand's technology creation performance has several explanations

New Zealand's low level of R&D output and lack of spillovers are both longstanding issues (OECD, 2007). Leading explanations for this relatively poor performance include:

- Economic structure – specifically the absence of big firms, geographic isolation and large primary sector (Crawford, Fabling, Grimes, & Bonner, 2007; Hong, Oxley, McCann, & Le, 2016; Jaffe, 2013).
- R&D and innovation policy spread too thinly across too many industries, without sufficient industry-government collaboration (Hendy, 2013; Skilling, 2020).
- A lack of innovation clusters, and of the large internationally-focussed frontier firms that support such clusters (NZPC, 2021).

Another potential explanation links New Zealand's technology creation challenges to its struggles with technology diffusion. Diffusion is, to some extent, complementary to technology creation, often preceding and enabling new innovations and discoveries (Breznitz, 2011; Ridley, 2020). Digital tools, for example, form a "key foundation for research and innovation" (OECD, 2020, p. 220), with international evidence showing a generally positive effect from firm adoption of digital tools on that firm's levels of R&D activity (Bello & Galindo-Rueda, 2020) and business innovation (Galindo-Rueda, Verger, & Ouellet, 2020). Many expect AI to support rapid scientific progress through more efficient R&D (Aghion & Bunel, 2024; OECD, 2023).

Overall, we judge New Zealand's low technology creation reflects a combination of multiple factors, including those highlighted in the various explanations above. A full analysis of these and other contending explanations drawn from the international literature (eg, absence of 'mission focussed' innovation (Mazzucato, 2021) or erosion of fundamental social institutions such as property rights (Moykr, 1992)) is outside this paper's scope. Yet each of the various explanations implies quite different things for innovation policy – including on funding, institutional design and sectoral concentration. In light of this and given the current fiscal position, we suggest, as others have previously (Jaffe, 2013), that getting the best out of current R&D expenditure remains a priority for New Zealand.

A set of principles for efficient R&D policy in a small open economy, which do not prejudge questions of funding, policy design or concentration, appear in the recent literature. These could, in theory, apply equally to a sector-focussed or broad-based innovation system. Specifically, existing studies recommend strong evaluation, careful policy design to avoid duplicative and distortionary incentives, and government focus on fundamental research (IMF, 2024; OECD, 2015). Strong framework conditions – particularly competitive markets, open trade settings and adequate finance – also support technology creation and the conversion of spillovers into productivity improvements (Bloom, Van Reenen, & Williams, 2019; Westmore, 2013).

Diffusion and adoption of technology may be particularly important for innovation in New Zealand

As discussed, New Zealand has low capital intensity and fairly low rates of technology creation. These factors may each be driving poor productivity performance. A third potential driver is the rate at which New Zealand firms adopt new technology. Technology adoption typically involves capital investment. There are five reasons diffusion and adoption via capital investment may be particularly important for New Zealand's productivity.

- 1 New Zealand conducts a tiny fraction of international R&D, with current expenditure 0.02% of the OECD total. Even if spending or efficiency grows sharply, New Zealand will always rely to a significant degree on adopting technology from overseas.
- 2 Adopting technologies is a more efficient path to innovation than creating them. As a result, diffusion strongly supports growth. Twenty five percent of the differential in countries' GDP per capita growth throughout the 20th Century is down to variation in how quickly and widely different countries adopt new innovations, according to one estimate (Comin & Hobijn, 2011).
- 3 Diffusion of foreign R&D imparts larger productivity improvements than domestic R&D, including for most OECD members, and particularly for small open economies similar to New Zealand, such as Ireland (Sakurai, Ioannidis, & Papaconstantinou, 1996; Santacreu, 2006; Santacreu, 2015). The New Zealand-based evidence is limited and somewhat ambiguous.¹³ On balance, and given its strong effects in the agricultural sector (Hall & Scobie, 2006), we judge that New Zealand does receive productivity benefits from foreign R&D.
- 4 Diffusion and adoption may become an increasingly attractive pathway to innovation. The race to reach the global technological frontier has strong winner-take-all dynamics.¹⁴ Many countries have tried and failed to reach the global frontier, sometimes at great cost (Breznitz, 2021; Taylor, 2016). Technology creation costs also appear to be increasing (Bloom, Jones, Van Reenen, & Webb, 2020) and possibly most sharply in small countries like New Zealand (Wakeman & Le, 2015).

¹³ Johnson et al. (2007) find no evidence of productivity benefits from foreign R&D in New Zealand. Their model suffers, however, from using Australian R&D spending as a proxy for all foreign R&D.

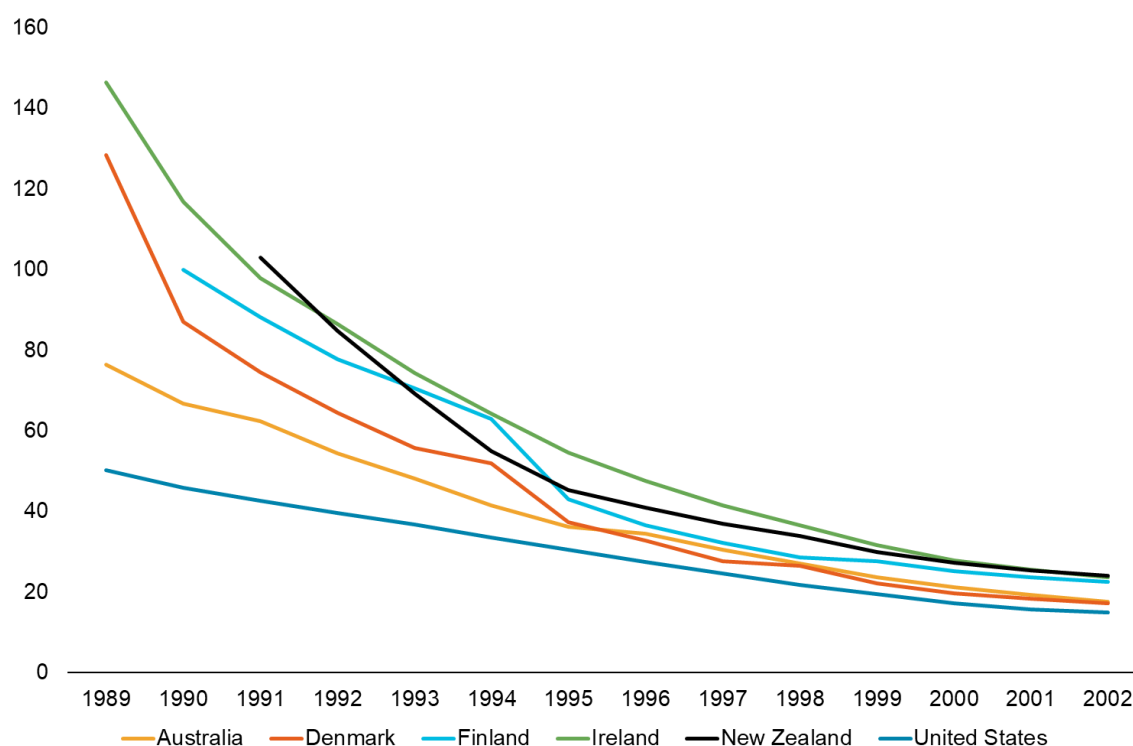
¹⁴ Among OECD countries more than 75% of total patents filed in 2023 were by firms in only five countries.

- 5 Diffusion affects a large proportion of New Zealand firms. According to Business Operations Survey (BOS) data from 2023 nearly half of reporting firms performed some type of innovation activity, a rate that has stayed steady for over a decade (Wakeman & Le, 2015). Fifteen to twenty percent say they introduced a new-to-firm product or process, and only a small proportion (approximately 5%) report creating new-to-world products or processes. That suggests around 95% of New Zealand firms are benefiting from, or could potentially benefit from, technology diffusion.

Dynamics of technology diffusion in New Zealand

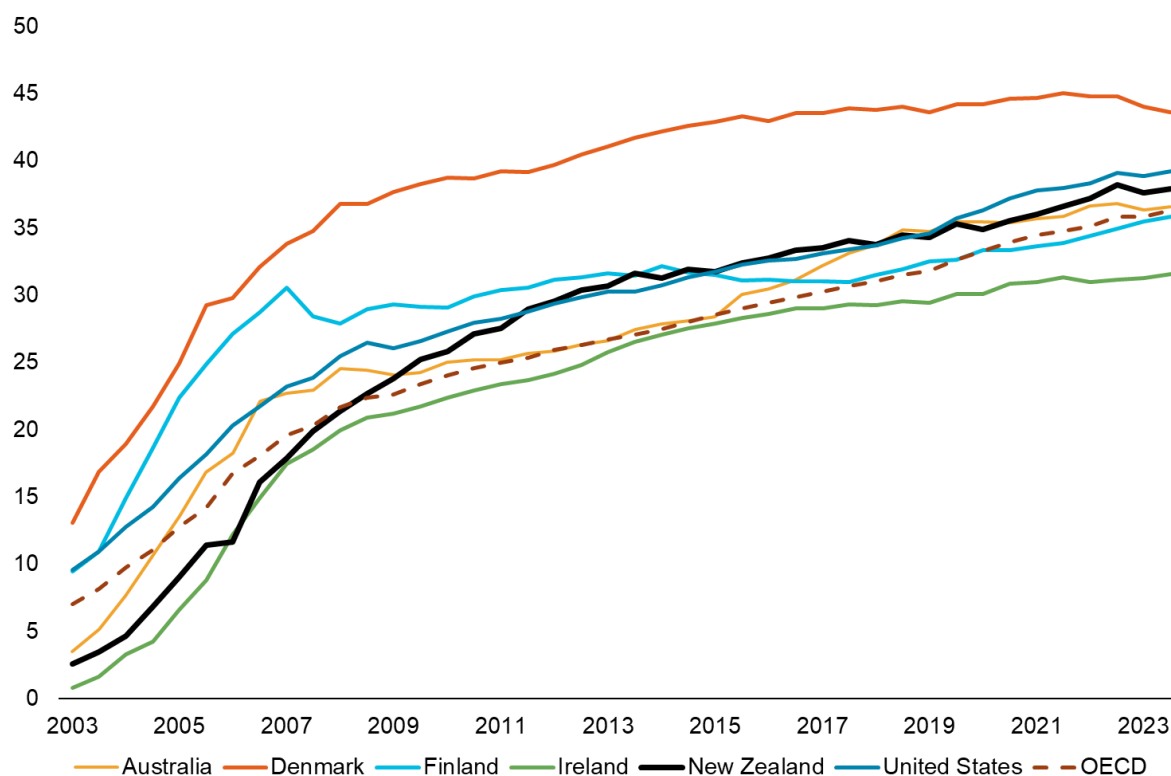
The evidence on technology diffusion in New Zealand is more circumstantial, more mixed and less fully corroborated than that on capital intensity and technology creation. As a result, our conclusions are tentative. Focussing only on two general-purpose technologies (GPTs) (computers and broadband internet) suggests New Zealand has kept pace with similar countries in terms of diffusion. (Figure 8 and Figure 9 – the US represents the global frontier.)

Figure 8: Penetration of computers, people per computer



Source: Cross-country Historical Adoption of Technology dataset, Comin & Hobijn, 2011.

Figure 9: Broadband penetration, subscriptions per 100 people



Source: OECD, Fixed Broadband Subscriptions.

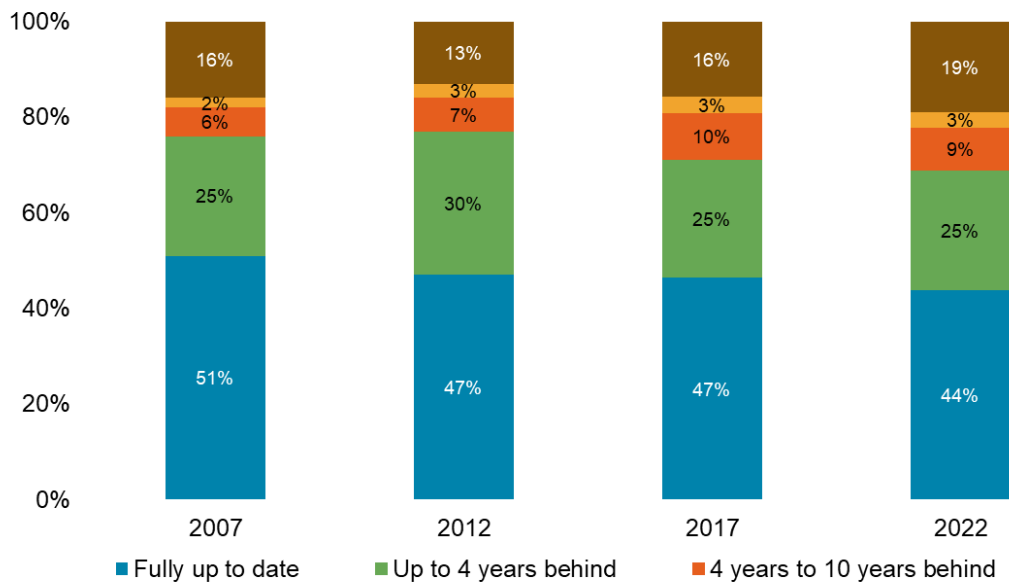
Note: Includes total number of subscriptions to broadband internet with download speeds of 256 kbit/s or greater.

The relatively rapid diffusion of computers and broadband in some senses defies expectations, particularly given New Zealand's geographic isolation. Yet other evidence suggests that levels of diffusion beyond computers and broadband are actually lower and slower than implied by Figures 8 and 9. The key metric in this regard is the sustained negative gap between the productivity of New Zealand's leading – "frontier" – firms and those in other SAEs. If effective technology diffusion was occurring, we would expect New Zealand frontier firms to converge to the SAE frontier. This has not happened (NZPC, 2021). Because frontier firms play a key role in receiving, and then adapting technology for local conditions, poor frontier firm performance could be impairing other firms' adoption of technology (Andrews, Criscuolo, & Gal, 2015).

One potential explanation for the apparent disconnect between the relatively rapid diffusion of broadband and computers, and low measured productivity lies in the possible failure to build out complementary intangible assets (Brynjolfsson, Rock, & Syverson, 2020). These assets, such as new business models and processes, are required to realise the potential of GPTs. National accounts can fail to measure such assets correctly, creating a "J-curve" effect where productivity initially dips as firms make complementary intangible investments before rising once the benefits of technology change are realised. Data limitations mean we cannot form a view on whether this effect explains the disconnect between New Zealand's GPT diffusion and productivity growth.

Approximately 70% of New Zealand firms self-assess that they operate within four years of their respective industries’ global technology horizon (Figure 10). At the same time, levels of technology among small firms have declined in recent years with fewer reporting upgrades to their core technologies. This supports the conclusion that diffusion in New Zealand is low and slow, and matters materially because domestic technology diffusion strongly influences wider economic productivity (Berlingieri, Calligaris, Sara, Criscuolo, & Verlhac, 2019).

Figure: 10 Average age of core technology over time

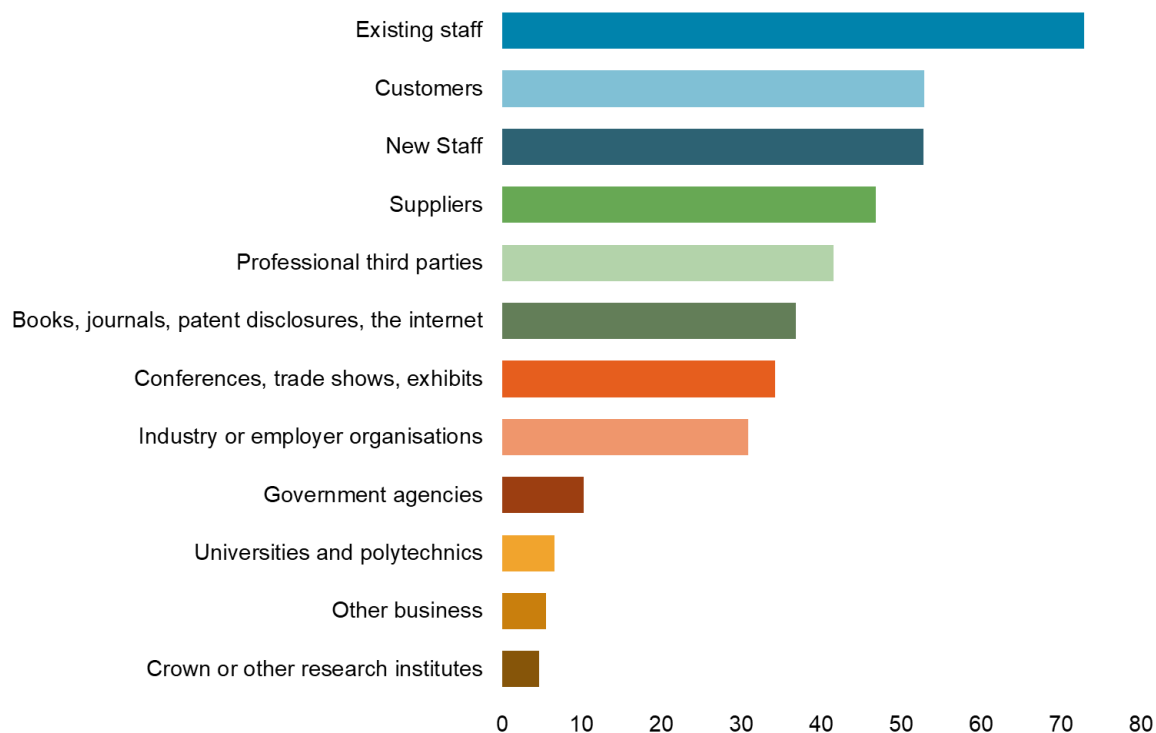


Source: Stats NZ Business Operations Surveys, 2007-2022 and Treasury calculations.

Information is scarce on the causal mechanisms driving diffusion in New Zealand. From productivity levels, we can infer that New Zealand firms adopt new innovations via their local regional frontier, rather than from the national or global frontier. Trading with other New Zealand regions can circumvent this localisation effect and promote convergence with the national frontier (Zheng, 2016). We also know that the productivity gap from the top 10% of firms to the bottom 10% has overall held constant since 2003 (Devine & Smith, 2025). This could imply a strong diffusion process – or, more likely in our judgment, a wide gulf from the New Zealand national frontier to the global frontier that allows lagging firms to keep up.

At a more detailed level, BOS data shows what types of technology firms adopt. In 2023 approximately 20% of New Zealand firms acquired computer software and hardware in order to pursue innovation. Fifteen percent acquired new machinery and equipment for innovation purposes, and 5% acquired new knowledge in the form of licenses and IP. BOS data also reveals the major sources of information that New Zealand firms rely on for innovation. As Figure 11 shows, staff and business relationships are the most frequent sources of information for innovations.

Figure 11: Sources of information for innovation, percent of reporting firms 2023



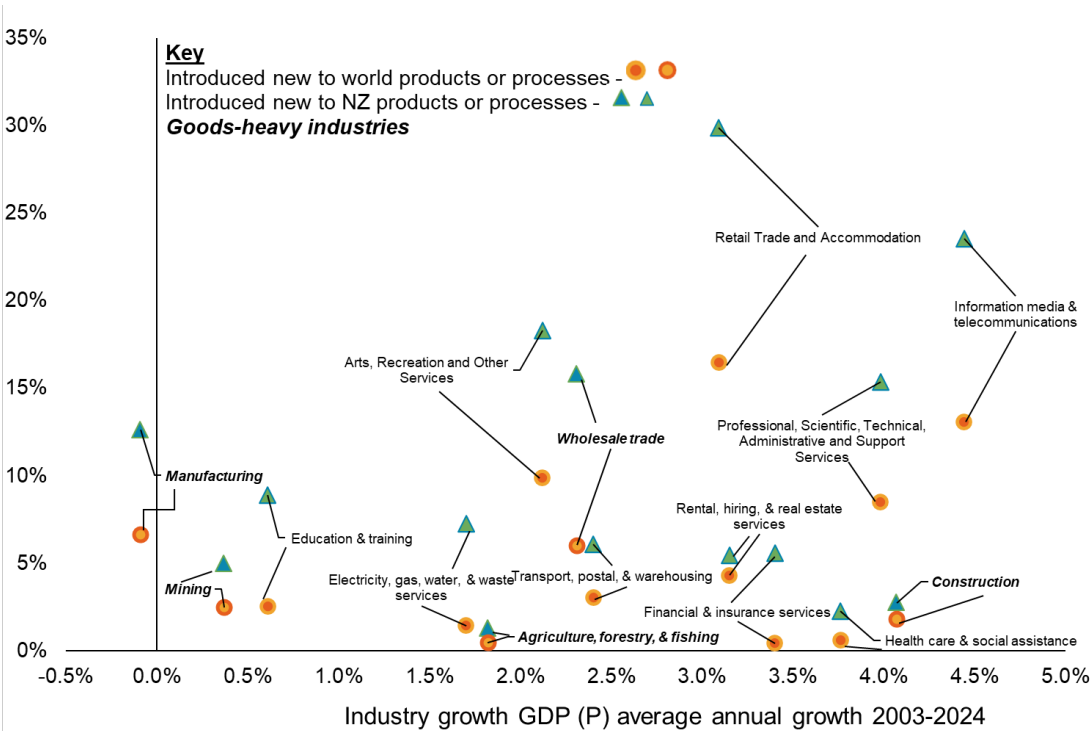
Source: Stats NZ Business Operations Survey, 2023.

An industry picture of technology diffusion in New Zealand

Industries rely on technology creation and diffusion to differing degrees, potentially reflecting their constituent firms' respective distances from the global frontier. Figure 12 presents the share of businesses involved in new-to-New Zealand innovation and new-to-world innovation relative to GDP growth in the past 20 years. From it we draw two initial conclusions:

- 1 The high growth and high innovation industries tend to be services. Though service industries all tend to exhibit relatively strong growth, there is a divergence in the levels of innovation across service industries. Goods-heavy industries tend towards either low growth and relatively high innovation (eg, manufacturing) or high growth and relatively low innovation (eg, construction). Overall, this suggests New Zealand's shifting industry composition towards services may also be affecting the nature and extent of innovation in the economy. (For example, innovation in services often involves less patenting than innovation in goods-heavy industries.)
- 2 A positive association seems to hold between industry reliance on technology adoption (new-to-New Zealand innovation) and technology creation (new-to-the world innovation). Firms in high-innovation industries seem to both adopt and create technologies. Firms in low-innovation industries apparently do less of both.

Figure 12: Innovation in New Zealand industries 2023, percent of reporting firms



Source: Stats NZ Business Operations Survey, 2023 and Treasury calculations.

4 Technology diffusion and links to capital

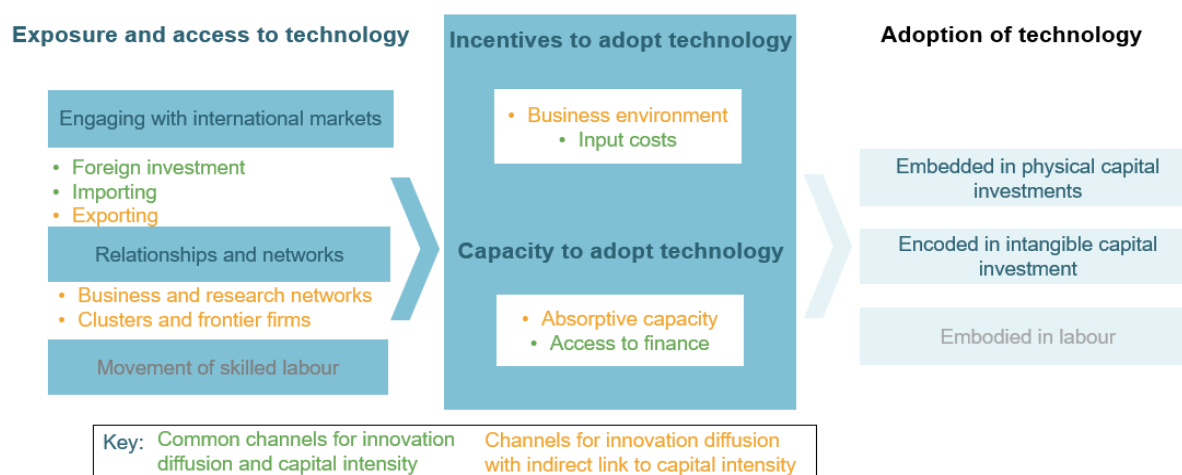
New Zealand's poor productivity performance is driven in part by its low levels of capital intensity and innovation, as laid out in sections 2 and 3. Technology diffusion, which often involves investment in more sophisticated capital, offers a way to increase both capital intensity and innovation. To better understand technology diffusion we present a framework (Figure 13) where innovation occurs via firms' adoption of technology embedded in physical capital and encoded in intangible capital. (We do not directly address labour and human capital issues.)

The framework aligns with similar OECD analyses (Andrews, Nicoletti, & Timiliotis, 2018). It is structured around three firm-level factors necessary for the adoption of new technology:

- exposure and access to new technology
- incentives to adopt new technology
- capacity to adopt new technology.

Each of these factors encompasses a number of channels that enable technology adoption. Technology adoption often occurs through investment in capital, meaning some channels overlap directly with capital intensification. We refer to these as “common channels”. Others with little overlap we refer to as having an “indirect link” to capital intensity.

Figure 13: Framework for technology diffusion



Source: The Treasury.

Our framework is not an exhaustive map of all the channels by which diffusion and capital intensification might occur. The focus is on what we take to be the most significant channels for diffusion and their connections to capital investment. There may be channels (and barriers) that are distinct for capital that have little to do with diffusion.¹⁵

¹⁵ For example, costs and uncertainties for construction as a result of planning and building regulation.

The framework omits some complexity in order to remain tractable. For example, foreign-owned firms often have access to different types of financing than domestically owned ones. FDI could therefore fall under ‘capacity’ as well as ‘access and exposure’. We rely on judgment to determine the ultimate position of such channels in the framework. In what follows we review the channels in terms of evidence of their effect on technology diffusion and links to capital intensity. Our findings are summarised in Table 1 below.

Table 1: Summary of channel effects and evidence

Channel	Effects on technology diffusion		Capital intensity link
	International evidence	New Zealand evidence	
Foreign investment	FDI involves private technology transfers and can lead to further diffusion among domestic firms Some evidence of reverse technology transfer from outbound investment	Private technology transfers occurring, mixed evidence on productivity spillovers from FDI Lack of evidence on effects of outbound investment on technology	Common channel: FDI can increase capital intensity if it goes beyond purchase of existing assets Outbound investment may divert capital from domestic stock
Importing	Strong evidence importing increases diffusion, including through competition effects	Very likely leads to diffusion in NZ despite only partial evidence	Common channel: Imports of capital, over and above replacement, can lift capital intensity
Exporting	Exporting prompts improvements to capital stock through firms upgrading technology	Suggestive NZ evidence that higher productivity firms tend to export	Indirect link: Exporting firms tend to invest more based on expected revenue
Business and research networks	Networks strongly support diffusion, especially global business relationships	Limited systematic evidence in NZ context	No connection
Clusters and frontier firms	Strong evidence that ideas and technology diffuse rapidly in clusters	Limited evidence in NZ – in part because of few clusters and frontier firms	Indirect link: Through infrastructure investment
Business environment	Competition and enabling regulation support diffusion	Limited evidence of effect of business environment on diffusion in NZ	Indirect link: Competition and enabling regulation support investment
Input costs	Higher capital input costs reduce investment in new technology	Anecdotes from some NZ sectors that high capital costs depress investment	Common channel: High capital costs can depress capital intensity
Absorptive capacity	Technical skills and absorptive capacity tend to increase diffusion	Strong management practices correlate with diffusion, staff leading source of innovation	Indirect link: Stronger management practices can increase capital intensity
Access to finance	May hamper technology adoption in small, young and intangible-heavy firms Large and sophisticated capital markets promote diffusion at a national level	May be having a negative impact on some firms’ ability to upgrade capital	Common channel: Greater access to finance can increase capital intensity

Exposure and access to new technology

Engaging with international markets

Foreign investment

Economic theory suggests that FDI improves domestic productivity as local subsidiaries adopt the competitive advantages of foreign parent companies (Dunning, 2000). These transferred advantages may include technology, though this depends on investor motivations and the character of the technologies in question, among other factors (Kogut & Zander, 2003; Sampson, 2024). In New Zealand, firms with foreign investors tend to be more productive, capital intensive, and return-generating than those without foreign investors. Technology diffusion may drive these outcomes, though foreign investors could simply be buying New Zealand's most productive firms (Fabling & Sanderson, 2014).

FDI can have wider economic effects, including stimulating greater technology adoption among other domestic firms that are not local subsidiaries (Blomström & Kokko, 1998). However, foreign firms have a strong incentive to prevent such spillovers to competitors, which erode their competitive advantage. Hence, domestic suppliers of foreign-owned firms, rather than the competitors of such firms, may receive the most spillover benefits from FDI (André & Gal, 2024).

Countries capture FDI spillovers most effectively where they have competitive and open domestic markets, and high absorptive capacity (Cohen & Levinthal, 1990; Javorcik, 2004; Leshner & Miroudot, 2008). New Zealand-based studies find little trace of economically material FDI spillovers in the aggregate. At a more granular level, small New Zealand firms and those in the primary sector appear to get a productivity increase from supplying foreign-owned firms (Doan, Maré, & Iyer, 2015). This may in part reflect diffusion through backward linkages – though the evidence is inconclusive.

In addition to its effects on technology diffusion, FDI can potentially increase the national capital stock. This effect depends in part on the allocation of any associated financial inflows across sectors, and between labour and capital.

How New Zealand's very low level of outbound investment affects technology diffusion is a gap in our knowledge. One study found New Zealand firms that invest offshore are more likely to engage in some types of innovation (Hong, Oxley, McCann, & Le, 2016). International evidence suggests outbound investment creates opportunities for reverse technology transfer, including through proximity to firms and inventors in innovative countries (Chen & Dauchy, 2018). Still, these effects depend greatly on contextual factors which may not apply to New Zealand (Amann & Virmani, 2014; Griffith, Harrison, & Van Reenen, 2005).

Importing

Consumers and firms benefit from access to foreign products, including technologies more sophisticated than those available domestically. In addition, international evidence shows importing supports technology diffusion by incentivising mimicking and stimulating competition among affected domestic markets (Bloom, Draca, & Van Reenen, 2016). Little research explores import-driven diffusion in New Zealand, though at least one historical study suggests it may have had a role in the development of the forestry sector (Roche, 2018). Regardless, the theoretical link from imports to technology diffusion remains compelling – and leads us to conclude that imports are a key channel for technology diffusion in New Zealand.

Imports also directly affect the national capital stock. New Zealand imports much of its physical capital. Because some elements of technology are tacit, firms find it easier to integrate imported capital assets into production where the vendor offers support, and operating instructions are highly codified (Kiriya, 2012; Kogut & Zander, 2003). Intangible imports (including IP licences and services) increase the capital stock as well, and also increase the set of ideas domestic innovators can access (IMF, 2024; Kiriya, 2012).

Exporting

Exporting can promote technology diffusion in two ways. First, exporting increases expected returns and can lead to exporters upgrading their technology and capital (Aw, Roberts, & Xu, 2011; Lileeva & Trefler, 2010; Verhoogen, 2008). Consistent with this international evidence, New Zealand firms entering overseas markets tend to experience an increase in productivity, capital investment and some types of innovation (Fabling & Sanderson, 2013; Ffowcs-Williams, 2009; Hong, Oxley, McCann, & Le, 2016). Exporting can thus have a positive knock-on effect for the capital stock. Second, exporting firms may pick up new innovations from more advanced and competitive overseas markets (De Loecker, 2007). This learning effect is present among New Zealand exporters, according to existing evidence (Fabling & Sanderson, 2013).

Relationships and networks

Business and research networks

Innovation is fundamentally a social activity (Ridley, 2020). Accordingly, the strength of relationships among businesses and with researchers in part determines a country's overall level of innovativeness. Relationships also influence diffusion by increasing firms' exposure and access to new products and processes. Business relationships with the greatest bearing on innovation are those associated with Global Value Chains (GVCs) that link together internationally-dispersed suppliers, users and customers (Pittaway, Robertson, Munir, Denyer, & Neely, 2004). A large literature explores the effects of GVCs on technology transfer and productivity. The findings are context specific, though broadly positive for technology diffusion and for productivity (Alfaro-Ureña, Manelici, & Vasquez, 2022; Sampson, 2024). Effective networks from public researchers to business can also spur innovation (Breschi, Lassébie, Lembcke, Menon, & Paunov, 2019).

New Zealand firms often rely on relationships with customers and suppliers for sources of new information (Figure 11). But New Zealand has low participation in GVCs and New Zealand firms rarely collaborate with higher education or research institutes (Deliotte, 2018), potentially because of cultural and institutional barriers (Science System Advisory Group, August 2024). One bright spot is the share of publications jointly authored with industry which has increased marginally of late, albeit from low levels (CWTS, 2025).

Government's potential role in developing innovation networks remains unclear, in part because past attempts were not subject to appropriate evaluation (Warwick & Nolan, 2014). National science systems can act as networks for diffusion – though in some past cases have inadvertently promoted the development of ultimately less-successful technologies over other more-successful ones (Pittaway, Robertson, Munir, Denyer, & Neely, 2004). Business accelerators and university incubators have had mixed success as well (Blair, Khan, & Iftikhar, 2020; Madaleno, Nathan, Overman, & Waights, 2022). Despite these issues, New Zealand's relatively low starting point may mean small improvements to innovation networks could have large positive effects.

Clusters and frontier firms

Clusters are local concentrations of innovative businesses in related or complementary fields. Clusters have similar features to networks, such as shared inputs (eg, via labour markets). Though unlike networks, clusters are bounded by geography (either at the city or country level), involve a mix of competition and co-operation, and do not typically restrict membership (Ormerod & Roswell, 2009; Porter & Stern, 2001; Warwick & Nolan, 2014). Clusters reflect the strong positive effect of proximity on innovation and diffusion (Breschi & Lissoni, 2009; Keller, 2002). This effect remains despite improvements to transportation and communication technology shrinking distance in other ways: firms still rely on innovations developed locally (Jaffe, Trajtenberg, & Henderson, 1993), innovate more often if located close to other innovative firms (Matray, 2021; Orlando, 2000), and invest more in intangibles if based in dynamic city clusters (Haskel & Westlake, 2022).

Proponents argue New Zealand should deliberately encourage clusters in specific sectors, as some other countries do. Yet the evidence on such policies is at best mixed. More impactful may be enabling 'organic' cluster formation. (Warwick & Nolan, 2014). As Haskel and Westlake (2022) show, dynamic cities support innovation and investment in intangibles. These cities require effective transport systems, high population density and job mobility. New Zealand therefore may benefit from enabling shifts in labour and construction regulation, and in infrastructure and planning. These shifts could supplement the country's already relatively strong digital infrastructure (Fabling & Grimes, 2016) and help overcome the geographic barriers to growth faced by some major and potentially highly innovative cities, including Auckland.

Large and globally-connected frontier firms may support cluster formation. These firms serve as "anchors" by increasing the commercialisation of local research and supporting younger firms – including by sharing innovations and attracting or training specialist workers (Agrawal & Cockburn, 2003; Ingstrup, 2014). Potentially because of its lack of very large firms in this anchor class, New Zealand exhibits limited clustering, even in areas of comparative advantage (NZPC, 2021; Skilling, 2020). As noted previously, some argue that this lack of very large firms dampens overall innovation in New Zealand.

Incentives to adopt new technology

Business environment

A competitive business environment can incentivise domestic firms to adopt new innovations, although competition's effects are not uniform across markets and contexts (Cherif & Hasanov, 2021). International evidence suggests innovation declines at extremely high and low levels of competition, and that innovation is higher where markets have low barriers to entry and where innovators judge they could feasibly reap profits from innovating (Aghion, Bloom, Blundell, Griffith, & Howitt., 2002). Though much of this evidence comes from patent activity, a consensus has emerged that similar dynamics likely affect diffusion (APC, 2023).

Given New Zealand's low levels of competition in some key sectors, it would almost certainly see innovation gains from greater competition (Dennerly, 2024). International evidence suggests competition can drive diffusion within sectors, such as agriculture, that are central to the New Zealand economy (Campi, Asai, McFadden, Pindado, & Rosburg, 2024).

In terms of capital intensity, relative competitiveness may also affect how capital is allocated across the economy. Firms may allocate capital towards areas of low competition where they can extract rents. Such allocation is unlikely to be socially optimal.

Regulation and regulatory barriers may also prevent investment in capital and limit firms' incentives or ability to invest in new technology. Regulation can increase uncertainty in investment decisions through delayed or unpredictable outcomes, and limit firms' risk appetites and flexibility (Murphy, Siedschlag, & McQuinn, 2012). Complex or overly restrictive regulations may require firms to allocate resources to meet compliance requirements, diverting funds away from innovation. Lack of regulatory certainty can also discourage long-term investment, as firms may hesitate to commit resources. At the same time, overly rigid regulation or the absence of enabling regulation can stifle innovation by constraining firms' ability to experiment with new technologies, business models, or processes that fall outside the scope of existing frameworks (Aghion, Bergeaud, & Van Reenen, 2023; Blind, 2012). This can be particularly challenging for smaller firms, which often lack the resources to navigate complex regulation.

Input costs

Input costs influence how firms allocate their resources. In the medium and long term, high labour costs motivate investment in more labour-saving technologies (Acemoglu, 2010). High capital costs also discourage investments and studies show this is particularly acute for both technology investment and, separately, for inbound investment (Clemens, Lewis, & Postel, 2018; Gallardo, 2018; Inland Revenue, 2022). Other factors also play a role, including uncertainty about a technology's local suitability, performance and reliability (Graf-Vlachy, Buhtz, & König, 2018).

The economy-wide implications of New Zealand's relatively high cost of capital for technology diffusion have not been empirically tested, though anecdotal insights exist. For example, horticultural businesses report that the relatively low cost of labour (combined with seasonal dynamics) makes it challenging to achieve an acceptable rate of return on investments in new technology (NZPC, 2021).

Capacity to adopt new technology

Absorptive capacity

Technical skills and management quality are key determinants of a firm's level of absorptive capacity. By technical skills we mean the level of specialised expertise (tacit and formal) available to a firm; the kind of expertise typically possessed by R&D workers. There is robust evidence showing firm investment in R&D (which often goes to personnel costs) increases technology adoption, including by reducing firm search and adaptation costs (Griffith, Redding, & Van Reenen, 2004). The Treasury has noted this dual role of R&D (promoting both creation and adoption) in the past (Blakely, Lewis, & Mills, 2005).

Some specific skills also increase adoption rates for particular technologies. For example, “STEM” skills support adoption of IT and AI (Draca, et al., 2017). More generally, foundational education remains key for broad-based and effective adoption and utilisation of new technologies (Riddell & Song, 2017).

Management quality also matters for technology diffusion. Technological change often entails adaptation and re-organisation among firms. Better management can enhance these processes and means firms ultimately make more effective use of new technology and capital (Andrews, Criscuolo, & Gal, 2015; Bloom, Genakos, Sadun, & Van Reenen, 2012; Giorcelli, 2019). These findings translate to New Zealand, where firms with greater absorptive capacity innovate and undertake R&D more often (Harris & Le, 2015). New Zealand firms using modern management practices also have greater productivity (Fabling & Grimes, 2009), and more frequently adopt new technologies (Green & Agarwal, 2011).

Access to finance

New innovations are inherently uncertain prospects. Technology adoption is typically a more reliable and less expensive innovation strategy than creation. But it is not risk free. As a result, firms wanting to invest in new technologies can face financial barriers. The increasingly intangible nature of technology is an exacerbating factor, as lenders often favour physical assets as collateral (Haskel & Westlake, 2022).

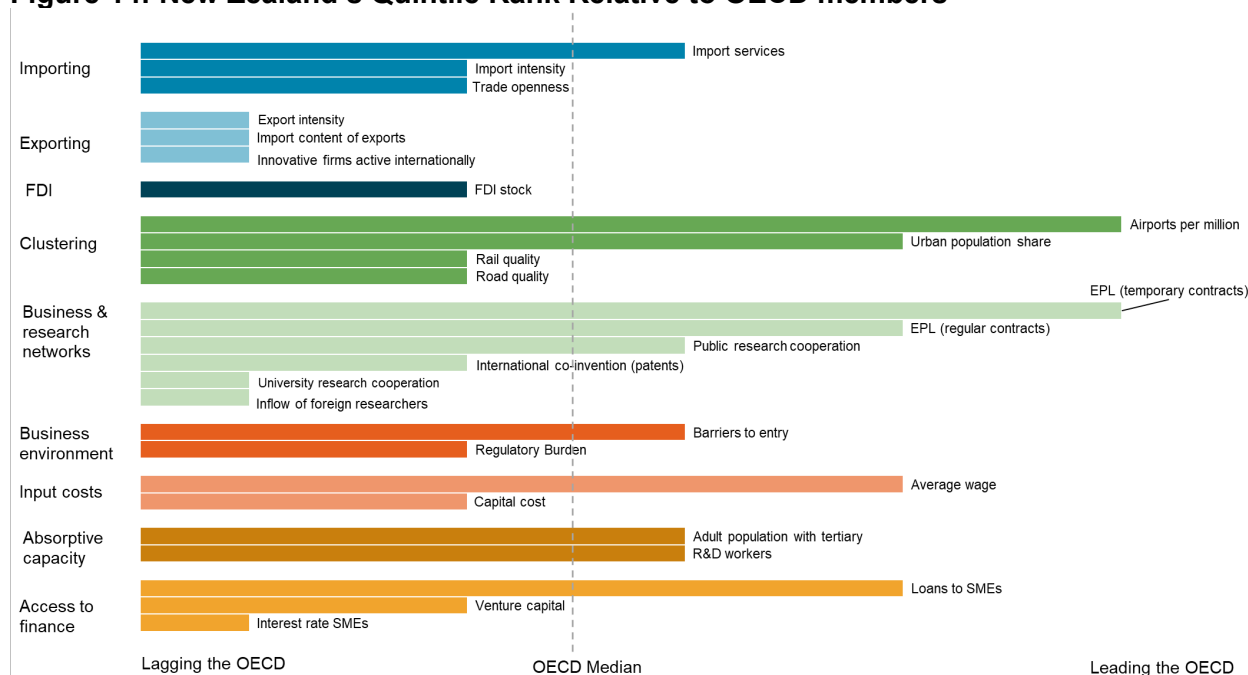
At the national level, high saving – channelled through mature capital markets – can support technology diffusion and long-term growth (Aghion, Howitt, & Mayer-Foulkes, 2004). No consensus exists on the optimal mix of capital market instruments (venture capital, angel investors, etc.) to enable technology diffusion (Breznitz, 2021; Kortum & Lerner, 2001). More important than the mix of instruments is the market's overall appetite to fund risky innovation prospects, particularly in young and small firms.

Observers, including the Treasury, have connected New Zealand's shallow capital markets with weak innovation outcomes (IMF, 2025; Treasury & MBIE, 2019). For small and young firms with low collateral, financial barriers probably do curb innovation, including investments in new technology. Māori firms may face similar struggles related to their unique characteristics, such as difficulty leveraging Māori freehold land for debt financing (Cherry & Cheung, 2025). BOS data suggest most New Zealand firms do not consider access to finance a material hurdle to expansion (NZPC, 2024). By our estimation, at most 5% of business capital investment plans in New Zealand in 2022 were stopped because the firm could not access finance on acceptable terms. In addition, relatively mobile global capital means access to finance is also not a major barrier to increasing the capital stock.

Strength of technology diffusion channels compared to OECD

Here we offer an initial assessment of the relative strength of New Zealand's technology diffusion channels compared to the OECD, ranking New Zealand's channel strength against the OECD median and quintiles (Figure 14). We perform this analysis by using indicators recommended in the OECD Innovation Diffusion Toolkit (Weingarden & Lembcke, 2024), with modifications to account for the (non)availability of New Zealand data and some additions so as to include indicators for all technology diffusion channels discussed above. (Further detail on indicators available in the Annex.)

Figure 14: New Zealand's Quintile Rank Relative to OECD members



Source: OECD, Stats NZ.

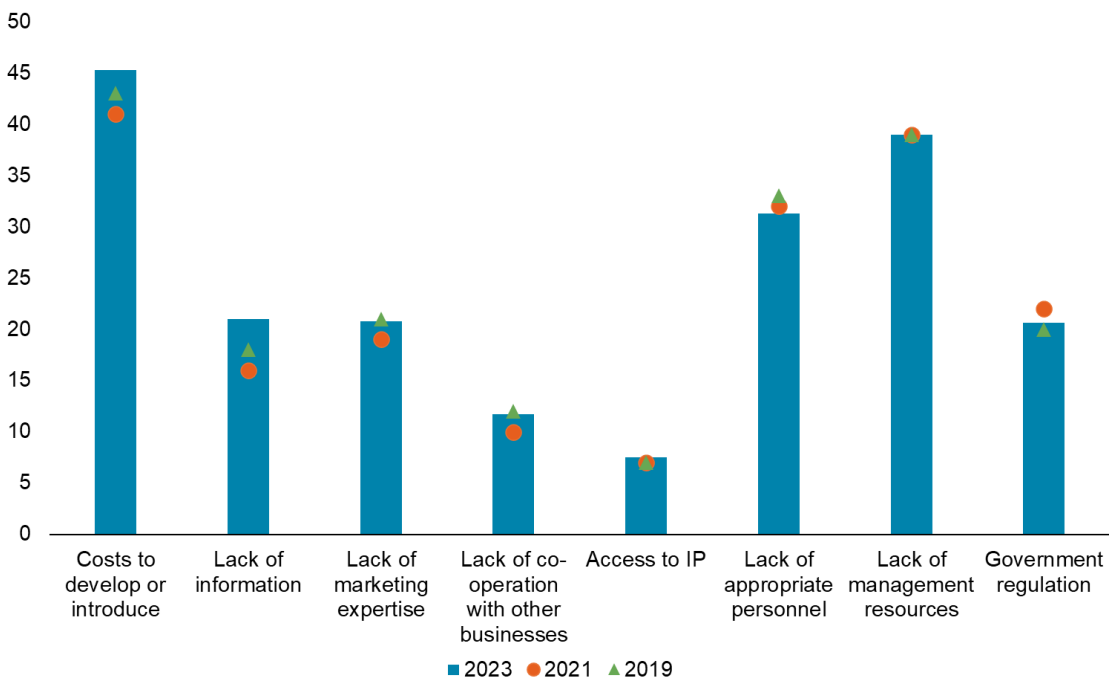
Note: The chart shows New Zealand's position relative to the OECD median across technology diffusion channels, with each channel represented by one or more indicator. Indicators all published between 2019 and 2024. Most indicator data from the OECD. EPL refers to Employment Protection Legislation strength, SMEs to Small and Medium Enterprises.

From the analysis we draw four conclusions:

- 1 Consistent with our overarching argument, many of New Zealand's weaker diffusion channels are common channels with capital intensity. Indicators associated with importing (import intensity and trade openness), FDI (FDI stock), clustering (infrastructure quality), and input costs (capital cost and interest rates to SMEs) all fall below the OECD median. We infer that lifting the strength of these common channels may yield a double benefit for productivity by increasing innovation through technology diffusion, and increasing capital intensity.
- 2 New Zealand's engagement with international markets strongly lags the OECD, according to exporting, importing and FDI indicators. Exporting is a particular weakness – including the import content of exports (a proxy for backward participation in GVCs).¹⁶ Further increasing export intensity may be challenging given New Zealand's distance to markets, though non-tourism service exports could offer one pathway. While New Zealand's imports of services are quite strong, imports of IP licenses and patents are low. As a result, New Zealand may not reap the technology diffusion benefits from its service imports implied by international evidence (IMF, 2024).
- 3 New Zealand is well positioned to take advantage of network and clustering effects owing to its strong public research cooperation, flexible labour market, and relatively high urban population. Addressing some lagging indicators could encourage deeper and more successful networks and clusters to form. Greater university-industry research cooperation, more international co-invention, and inflow of foreign researchers could be beneficial – though the New Zealand evidence on this last issue is somewhat ambiguous (Mare, Fabling & Stillman, 2011). Sufficient population density and quality infrastructure are also critical to unlock the benefits of a high urban population. Infrastructure is only captured crudely in Figure 14 in terms of airport numbers, and assessed road and rail quality. Population density is not included in the analysis.
- 4 The aggregate strength of technology diffusion channels as shown in Figure 14 does not correspond exactly with the barriers to innovation that New Zealand firms report facing. These barriers are summarised in Figure 15. New Zealand's relatively weak performance on channels associated with SME interest rates, venture capital, and capital cost is congruent with firms reporting cost as a major barrier to innovation. Low management capability has been identified as an issue in New Zealand previously (Green & Agarwal, 2011) and comes through in firm views of barriers to innovation. But it is difficult to measure accurately so not included in Figure 14. More work is needed to understand how the aggregate channels and the barriers to innovation New Zealand firms report facing map onto each other.

¹⁶ Exporting is not included in the four weak channels that are common to technology diffusion and capital intensity because (as described above) the link between exporting and capital intensity is indirect.

Figure 15: Barriers to innovation in New Zealand, percent of reporting firms 2023



Source: Stats NZ Business Operations Survey, 2023.

5 Areas for future research

Based on gaps identified in this paper, we suggest seven areas for future research. Filling these gaps would increase our understanding of the innovation process in New Zealand and ultimately lead to stronger policy design.

- 1 Geographic isolation may pose less of a barrier for the adoption of AI than it did for other GPTs, since fewer imported physical assets are likely to be required for AI. Future research that tracks AI adoption rates in New Zealand and compares it with the adoption of broadband, for example, could improve understanding of how distance influences the diffusion of technologies with different physical asset requirements.
- 2 Future research could assess innovation and growth at the firm level. Figure 12 offers a starting point for this research insofar as it presents an industry-level view of innovation and growth. The experiences of firms behind the national technology frontier are of particular significance in relation to technology diffusion. These firms have not had the same analytic focus to date as frontier firms, making policy design around these firms more difficult.
- 3 Services play an increasingly large role in the New Zealand economy and future research may explore what this means for innovation. One analytic puzzle that could motivate the work is the seeming divergence in innovation activity among service industries as presented in Figure 12.
- 4 R&D supports both the creation and diffusion of technology. But existing evaluations of R&D tend to focus on the benefits that accrue primarily from technology creation. Future research could assist policy design by analysing the benefits associated with technology diffusion through R&D.
- 5 New Zealand has very low outbound investment. Further work is required to understand what drives this, and what the implications are from a technology diffusion, and broader innovation, perspective.
- 6 This note largely avoided issues relating to labour and human capital, and their intersections with innovation. These remain potentially productive areas of inquiry. Among other topics, potential future research could re-examine existing work on the effects of people flows on innovation (Maré, Fabling, & Stillman, 2011).
- 7 Figure 14 presents an initial strength assessment of New Zealand's channels for technology diffusion. Future research could assess how material the effects of the various channels are for technology diffusion, capital investment and productivity growth respectively. Such work would be likely to shed light on some of the barriers to technology diffusion in New Zealand – which policy could then target.

Annex

The below table identifies the data used in Figure 14.

Winter, Devine, Janssen, Thompson (2025)	Indicator, source (date)	Source motivating indicator inclusion
Import services	Service imports, % GDP, OECD (2022)	IMF (2024)
Import intensity	Imports, % GDP, OECD (2022)	(Bloom, Draca, & Van Reenen, 2016)
Trade openness	Total trade, % GDP, OECD (2022)	(Weingarden & Lembcke, 2024)
Export intensity	Exports, % GDP, OECD (2022)	(Weingarden & Lembcke, 2024)
Import content of exports	Domestic value added, OECD (2020)	(Weingarden & Lembcke, 2024)
Innovative firms active internationally	Innovative firms active in foreign markets % of innovative firms, OECD (2020)	(Bloch & López-Bassols, 2009)
FDI stock	FDI % GDP, OECD (2022)	IMF (2024)
Airports per million	Airfields per million people, World Factbook (2021)	OECD (2024)
Urban population share	Proportion of city inhabitants, World Bank (2024)	(Haskel & Westlake, 2022)
Rail quality	Rail quality, OECD (2020)	(Weingarden & Lembcke, 2024)
Road quality	Road quality, OECD (2020)	(Weingarden & Lembcke, 2024)
EPL (temporary and regular contracts)	Strictness of employment protection, OECD (2019)	(Murphy, Siedschlag, & McQuinn, 2012)
Public research cooperation	Firms co-operating with higher education institutions, % innovative firms, OECD (2020)	(Weingarden & Lembcke, 2024)
International co-invention	International co-invention, % of PCT patent applications, OECD (2020)	(Weingarden & Lembcke, 2024)
University research cooperation	International collaborations on scientific publications, OECD (2023)	(Bloch & López-Bassols, 2009)
Inflow of foreign researchers	R&D immigrants as a share of population, OECD (2023)	(Bloch & López-Bassols, 2009)
Barriers to entry and regulatory burden	Product Market Regulation, OECD (2023)	(Weingarden & Lembcke, 2024)
Average wage	Average annual wages, OECD (2023)	(Acemoglu, 2010)
Adult population with tertiary	Completed tertiary education % of population, OECD (2022)	(Weingarden & Lembcke, 2024)
R&D workers	R&D personnel per 1000 workers, OECD (2023)	(Weingarden & Lembcke, 2024)
Loans to SMEs	Total loans outstanding % to SMEs, OECD (2021)	(Weingarden & Lembcke, 2024)
Venture capital	Venture capital, % GDP (2020)	(Bloch & López-Bassols, 2009)
Interest rate SMEs	Difference in the interest rate on loans between large and small firms (2022)	(Weingarden & Lembcke, 2024)

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